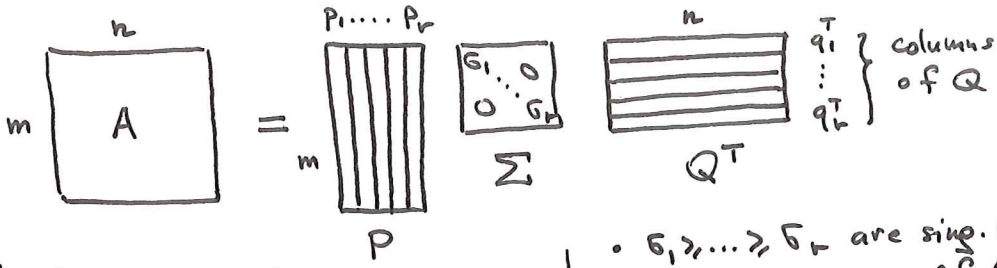
Lecture 17

Applications of SVD

- Image Compression
- Best rank-k approximation
- ~~Maximum Cut Problem~~ (Example from political science)
- Facial Recognition

Recall the SVD: any $A \in \mathbb{M}_{m\times n}$ $(A \neq 0)$



- $\sigma_1 \geq \dots \geq \sigma_r$ are sing. val. of A
- $r = \text{rank}(A)$
- $P^TP = I_r$
  $\{p_1 \dots p_r\}$ is an orthonorm. basis in $\text{im}\, A = \text{span}\{a_1 \dots a_n\}$
- $Q^TQ = I_r$
  $\{q_1 \dots q_r\}$ is an orthonormal basis in $\text{coim}\, A = \text{span}\{a^1 \dots a^m\}$

The key property that makes the SVD very popular in applications is its <u>numerical stability</u>.

More details on this in ACM 106, but num. stab. stems from the fact that P and Q have orthonormal colums,

$\Rightarrow$ do not change the length of a vector:

$$\|Px\|^2 = (Px)^T(Px) = x^TP^TPx = x^Tx = \|x\|^2.$$

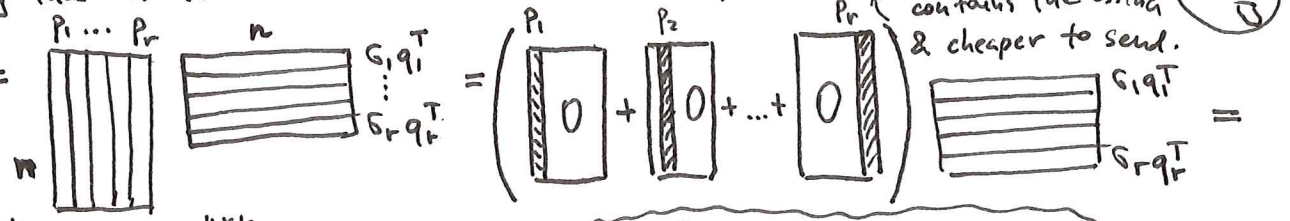
① Image Compression.

Suppose a satellite takes a picture and wants to send it to Earth.

Let A be the pixel intensity matrix: a_{ij} is the intensity of pixel (i,j)

Then transmission of A requires transmitting n^2 real numbers.

But often only the essential information inside A is required, $\tilde{A} \approx A$ contains the essence & cheaper to send.



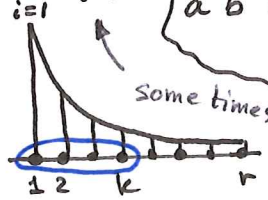
$$= \underbrace{p_1\sigma_1 q_1^T}_{\text{rank}=1} + \underbrace{p_2\sigma_2 q_2^T}_{\text{rank } 1} + \dots + \underbrace{p_r\sigma_r q_r^T}_{\text{rank } 1} = \sum_{i=1}^{r} \sigma_i p_i q_i^T$$

$n\times n$ $\quad$ $n\times n$ $\quad$ $n\times n$

$$a\,b^T = \begin{bmatrix} a \end{bmatrix} \begin{bmatrix} b^T \end{bmatrix} = \begin{bmatrix} b_1 a & \dots & b_n a \end{bmatrix}$$

rank 1

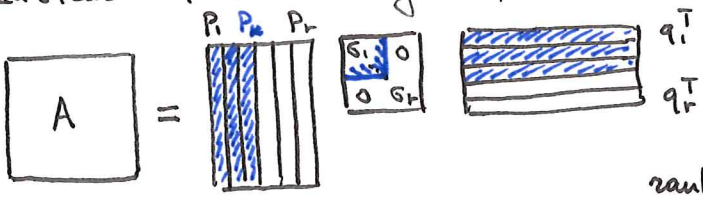
Often: $\underbrace{\sigma_1 \geq \sigma_2 \geq}_{\text{large}} \dots \underbrace{\geq \sigma_{r-1} \geq \sigma_r}_{\text{small}}$



Some times referred to as the dyadic decomposition

Idea:

Instead of transmiting A, lets transmit $A_k = \underbrace{\sum_{i=1}^{k} \sigma_i p_i q_i^T}_{\text{"thin" SVD of } A\text{, } k\text{-mode truncation.}} \approx A$ discarding small singular values



$\approx A_k$

$\text{rank}(A_k) = k$

Transmition A_k requires transmitting

$$kn + k + kn = k(2n+1) = O(kn)$$

If $k \ll n \Rightarrow$ savings

true for many images

Matlab Example

Here we replaced A of rank r with its approximation A_k of rank k.

Is this good approximation? May be $\exists\, B_k$ of rank k such that $A \approx B_k$ is better.

② Best rank-k approximation.

To compare different approximations, we need a measure of how close a given matrix to zero matrix. That is we need a matrix norm.

There are two important matrix norms: the 2-norm, $\|A\|_2$
the Frobenius norm, $\|A\|_F$. Take out

Def The 2-norm of a matrix A is given by $\|A\|_2 = \max_{x \in S^1} \|Ax\| = \sigma_1(A)$

Def The Frobenius norm of A is given by $\|A\|_F = \sqrt{\sum_{i,j=1}^{m,n} a_{ij}^2}$. $\left(= \sqrt{tr(A^TA)}\right)$

It turns out that the Frobenious norm is fully defined by the singular values of A.

$A = P\Sigma Q^T$ $\quad q_1, \dots q_r$ form an orthonormal basis in the row space of A.

Recall that (p.44) Lecture 10: if $u_1 \dots u_n$ is an orthonormal basis of V and $v \in V$

$$\Rightarrow \|v\|^2 = \sum \langle v, u_i \rangle^2$$

Let a^i be the ith row of A, then

$$\|a^i\|^2 = \sum_{j=1}^{r} (a^i \cdot q_j)^2$$

$$\|A\|_F^2 = \sum_{i,j} a_{ij}^2 = \sum_{i=1}^{m} \|a^i\|^2 = \sum_{i=1}^{m} \sum_{j=1}^{r} (a^i \cdot q_j)^2 = \sum_{j=1}^{r} \underbrace{\sum_{i=1}^{m} (a^i \cdot q_j)^2}_{\|Aq_j\|^2} = \sum_{j=1}^{r} \underbrace{\|Aq_j\|^2}_{\sigma_j^2(A)}$$

recall optimization interpretation of sing. values and vectors

$$\Rightarrow \boxed{\|A\|_F = \sqrt{\sum_{i=1}^{r} \sigma_i^2(A)}}$$

Or, it can be shown directly:

$$\|Aq_j\|^2 = (Aq_j)^T A q_j = q_j^T Q \Sigma P^T P \Sigma Q^T q_j$$

$$= q_j^T Q \Sigma^2 Q^T q_j = \sigma_j^2(A).$$

$$\begin{bmatrix} q_j^T \end{bmatrix} \cdot \begin{bmatrix} q_1 & \cdots & q_r \end{bmatrix} = [0 \dots \overset{j}{1} 0 \dots 0]$$

(Eckart & Young, [redacted]) 1936

Theorem: Let $A = \sum_{i=1}^{r} \sigma_i p_i q_i^T$ be the SVD of A, and let $A_k = \sum_{i=1}^{k} \sigma_i p_i q_i^T$, $k = 1, \dots, r$.

For any matrix B (of the same size), rank $B \le k$

$$\|A - A_k\|_F \le \|A - B\|_F$$

$$\|A - A_k\|_2 \le \|A - B\|_2$$

(spectral norm)

This result contributes a lot to the popularity of SVD.

1) Memory savings (like in image compression)
2) Eff. alg. for low rank matrices
3) Knoledge extraction from data,

SVD for data mining

political science example

Remark In fact, A_k is the best approximation to A for any unitary invariant norm (Mirsky, 1960), for example for $\|\cdot\|_2$

($\|UAV\| = \|A\|$ for any U, V unitary)

Ref [1] C. Eckart & G. Young, The approximation of one matrix by another of lower rank
([redacted]) Psychometrika, 1(3), 211-218

③ Application of the SVD to Political Science: Analysis of Senators' voting records

Suppose we have the following data:

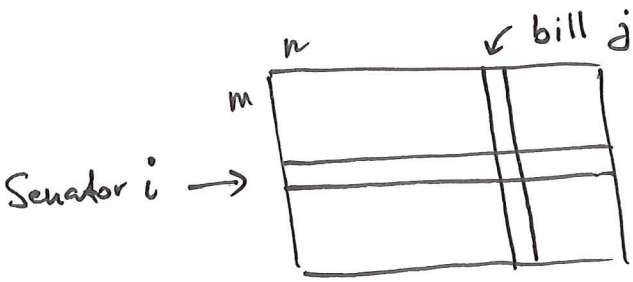
1) List of m senators

2) Vector $p = (p_1 \dots p_m)$ that encodes party affiliation

$$p_i = \begin{cases} +1 & \text{if Senator } i \text{ is Democrat} \\ 0 & \text{if Senator } i \text{ is Independent} \\ -1 & \text{if Senator } i \text{ is Republican} \end{cases}$$

3) Voting $m \times n$ matrix A that encodes how m senators voted on n bills.

$$A_{ij} = \begin{cases} 1 & \text{Senator } i \text{ voted for bill } j \\ 0 & \text{Senator } i \text{ did not vote for bill } j \\ -1 & \text{Senator } i \text{ voted against bill } j \end{cases}$$



Eeach Senator can be represented by a vector $A^i \in \mathbb{R}^n$ coim $A \subset \mathbb{R}^n$ (large)

Consider the SVD of A:

$$A = P \Sigma Q^T$$

We know that $\underbrace{q_1^T \dots q_r^T}_{\text{rows of } Q^T}$ form a basis of coim A

$$\Rightarrow A^i = \alpha_1^i q_1^T + \dots + \alpha_r^i q_r^T.$$

The problem is that $r \gg 1 \Rightarrow A^i \in \underbrace{\text{high dim}}_{\text{rank of } A}$ space $\Rightarrow$ difficult to visualize and make sense

Let's take a 2-mode truncation:

$$A \approx A_2 = \sigma_1 p_1 q_1^T + \sigma_2 p_2 q_2^T$$

Now each Senator i is represented by $(A_2)^i = \alpha_1^i q_1^T + \alpha_2^i q_2^T$

basis in coim A_2

To find coordinates α_1^i and α_2^i of Senator i, we need to solve the following linear system:

$$(A_2^i)^T = [q_1, q_2] \begin{bmatrix} \alpha_1^i \\ \alpha_2^i \end{bmatrix}$$

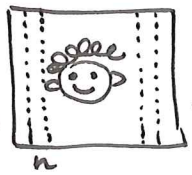
This way we will get a cloud of points in 2D, each representing a senator.

Let's look how it looks like for <u>Senate 107</u>

④ Eigenfaces for Facial Recognition

Problem: Given a set of facial images $\mathcal{F} = \{f_1, \dots, f_N\}$ (employees, medical patients) and a new image f, determine if $f \in \mathcal{F}$ and find the best match.

(Let's focus on this problem.)

$f =$ 

$\rightsquigarrow$ Let f be a facial image in a gray scale $f \in \mathbb{R}^{m \cdot n = D}$, even if the resolution is low ($m = n = 100$), the dim is very large.

Let $\mathcal{F}_{all}$ be set of all facial images.

Empirical result: $\dim \operatorname{span} \mathcal{F}_{all} \ll D$ $\quad \left(\dim \operatorname{span} \mathcal{F}_{all} \sim 10^2 \right)$ (independent on resolution!)

Assume that $\mathcal{F} \subset \mathcal{F}_{all} \subset \mathbb{R}^D$ represent $\mathcal{F}_{all}$ well: $\operatorname{span} \mathcal{F} \approx \operatorname{span} \mathcal{F}_{all}$.

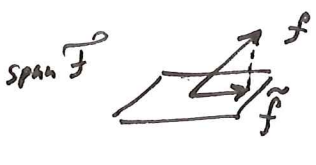
Idea: (Sirovich & Kirby, 1987)

(1) Find an orthonormal basis $p_1 \dots p_r$ in $\operatorname{span} \mathcal{F}$

$$\Rightarrow \quad f_i = \sum_{j=1}^{r} \alpha_j^{(i)} p_j \,, \qquad \alpha^{(i)} = (\alpha_1^{(i)} \dots \alpha_r^{(i)})$$

coordinate of image f_i in $\{p_1 \dots p_r\}$

(2) Represent a new image $f \in \mathcal{F}_{all}$ as



solve it in least sq. sense (or $\approx$)

$$f = \sum_{j=1}^{r} \beta_j p_j \qquad \beta^* = (\beta_1^* \dots \beta_r^*)$$

coord. of f in $\{p_1 \dots p_r\}$

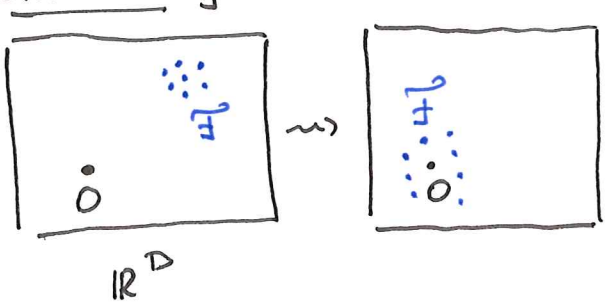
(3) Compare "features" β^* and $\alpha^{(1)}, \dots, \alpha^{(N)}$.

Implementation

(a) Assume that f_i are normilized, so that $\sum_{i=1}^{N} f_i = 0$

This is achieved by $f_i \rightsquigarrow f_i - \bar{f}$, $\bar{f} = \frac{1}{N} \sum_{i=1}^{N} f_i$

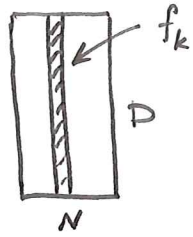
Geometrically:



Numerically: it is more efficient to work with normalized faces

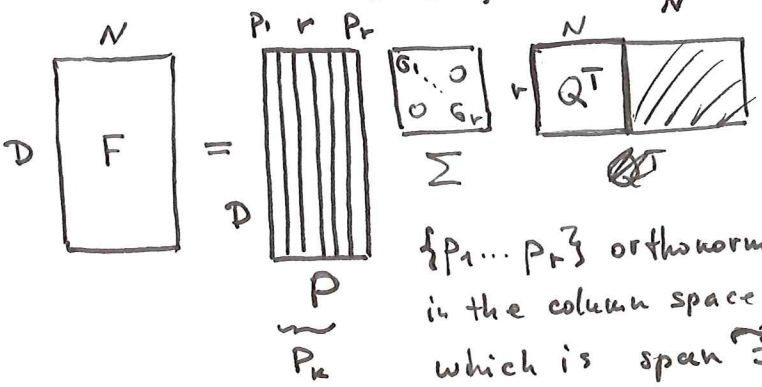
(singular values decay faster.)

(b) Let $F = [f_1 \dots f_N] =$



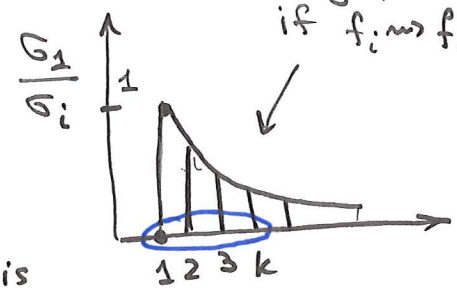
$D \gg N$

Let $F = P\Sigma Q^T$ be the SVD of F



$\{p_1 \dots p_r\}$ orthonormal basis in the column space of F, which is span $\mathcal{F}$.

decay faster if $f_i \rightsquigarrow f_i - \bar{f}$



Effectively, we can use $p_1 \dots p_k$ instead of $p_1 \dots p_r$.

Recall : columns of Q are singular vectors of F, that is eigenvectors of F^TF :

$$F^TF q_i = Q\Sigma \overbrace{P^TP}^{I} \Sigma Q^T q_i = Q\Sigma^2 \underbrace{Q^T q_i}_{\begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} i} = \sigma_i^2 q_i$$

Columns of P are eigenvectors of FF^T

$$FF^T p_i = P\Sigma Q^T Q \Sigma P^T p_i = \sigma_i^2 p_i$$

$\Rightarrow$ $p_1 \dots p_k$ are called eigen faces

(c) Using eigen faces $(p_1 \dots p_k)$

Given a new face f, we normalize it $f \rightsquigarrow f - \bar{f}$, and find the coordinates of $f - \bar{f}$ in $\{p_1 \dots p_k\}$:

$$\bar{f} = \frac{1}{N}\sum_{i=1}^{N} f_i$$

$$f - \bar{f} = \beta_1 p_1 + \dots + \beta_k p_k = P_k \beta \Rightarrow \boxed{\beta^* = P_k^T (f - \bar{f})}$$

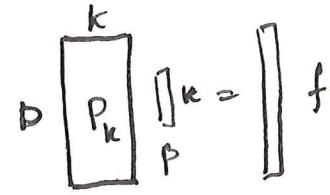
this is solution in the least squares sense

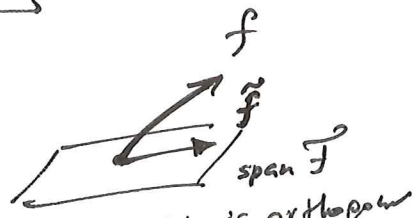
captures all features associated with f

To find the best match, we compare β with $\boxed{\alpha^{(i)} = P_k^T (f_i - \bar{f})}$

The eigen face reconstruction of f is $\boxed{\tilde{f} = P_k \beta^* + \bar{f}}$

$P_k \beta = f - \bar{f}$





$p_1 \dots p_k$ is orthogonal basis

unknown = k

equations D

$D \gg k$